

AI Resume Screener & Candidate Ranking Tool

Domain: HR Tech / Startup Tool

Team Members:

- Harshini S - Frontend (React UI)
- Konda Yashas Sree - Backend (FastAPI)
- Natarajan S - AI C Resume Processing
- Madduri Sai Mythili - PDF Parsing, Export C Integration

• **Mentor:**

R Abhinaya

Introduction

In today's competitive hiring environment, recruiters often receive a large number of resumes for a single job opening. Manually reviewing and shortlisting candidates is time-consuming, repetitive, and prone to inconsistencies. The AI Resume Screener C Candidate Ranking Tool leverages Artificial Intelligence to automate resume analysis, evaluate candidate suitability, and rank applicants based on job requirements, enabling faster and more efficient hiring decisions.

Problem Statement:

- Manual resume screening is time-consuming and inefficient.
 - Recruiters spend considerable effort reviewing large volumes of resumes.
 - Candidate evaluation can be inconsistent and subjective.
 - Startups and small businesses often lack dedicated HR teams.
 - Identifying the most suitable candidates from numerous applications is challenging.
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Proposed Solution

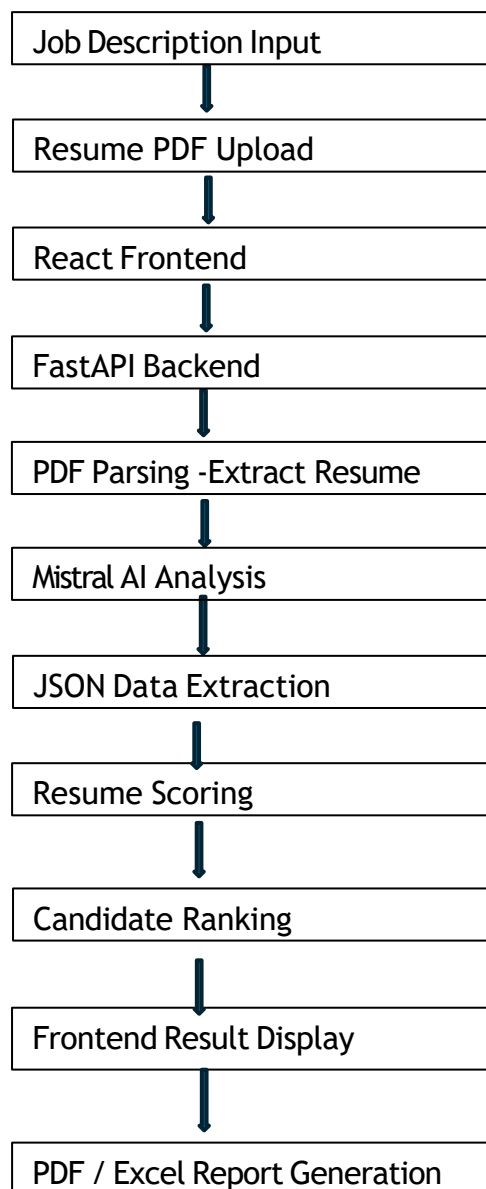
To address these challenges, An AI-powered web application that:

- ❖ Accepts job description from recruiter
 - ❖ Uploads multiple PDF resumes at once
 - ❖ Automatically extracts candidate information
 - ❖ Scores and ranks every candidate using AI
 - ❖ Generates exportable PDF and Excel reports
-

Objectives

- Accept job description text.
 - Upload multiple PDF resumes.
 - Extract skills, experience, education, and projects.
 - Compare resumes against job requirements.
 - Generate AI-based candidate scores.
 - Rank candidates automatically.
 - Compare candidates side-by-side.
 - Export reports in PDF and Excel formats.
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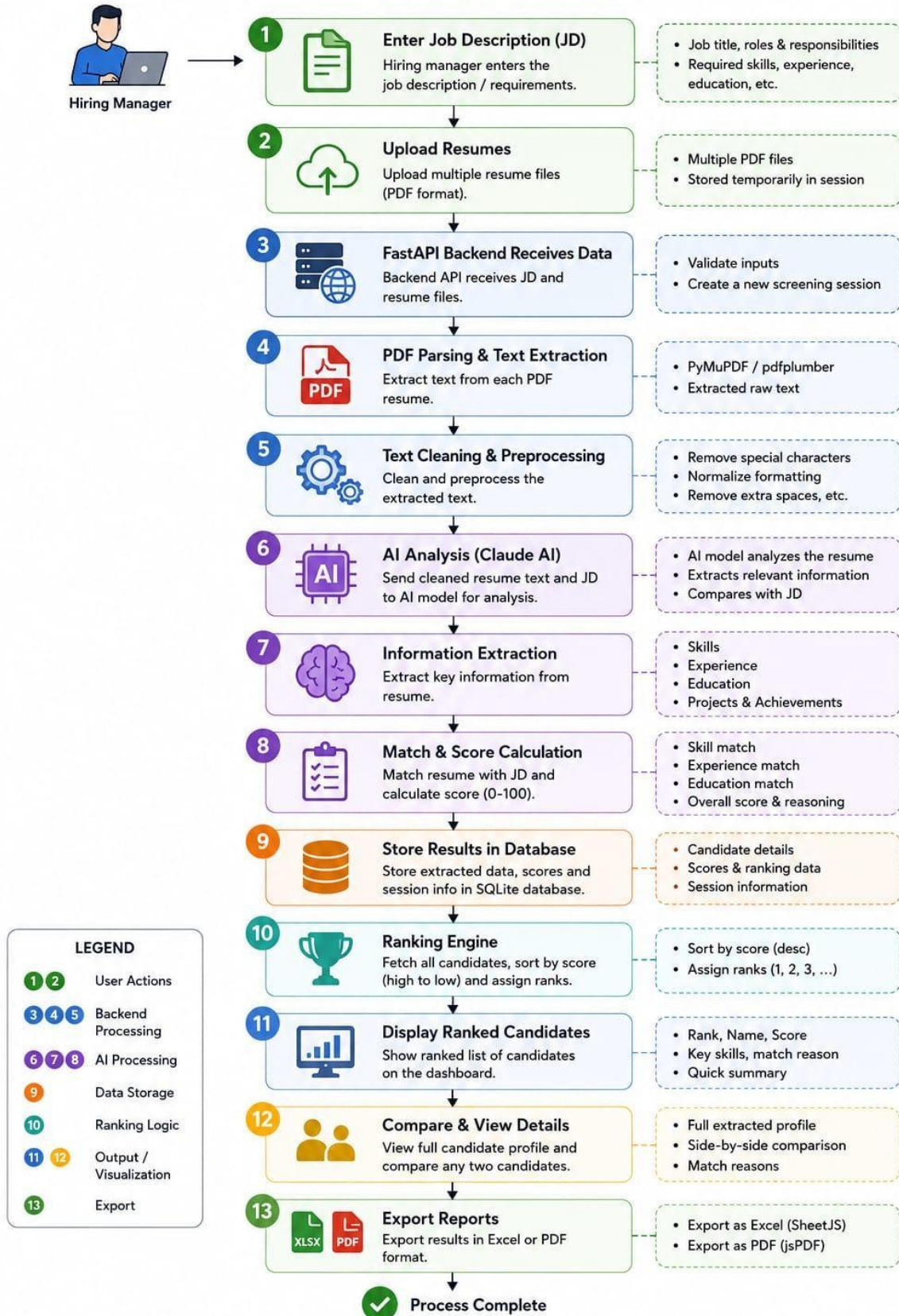
System Workflow



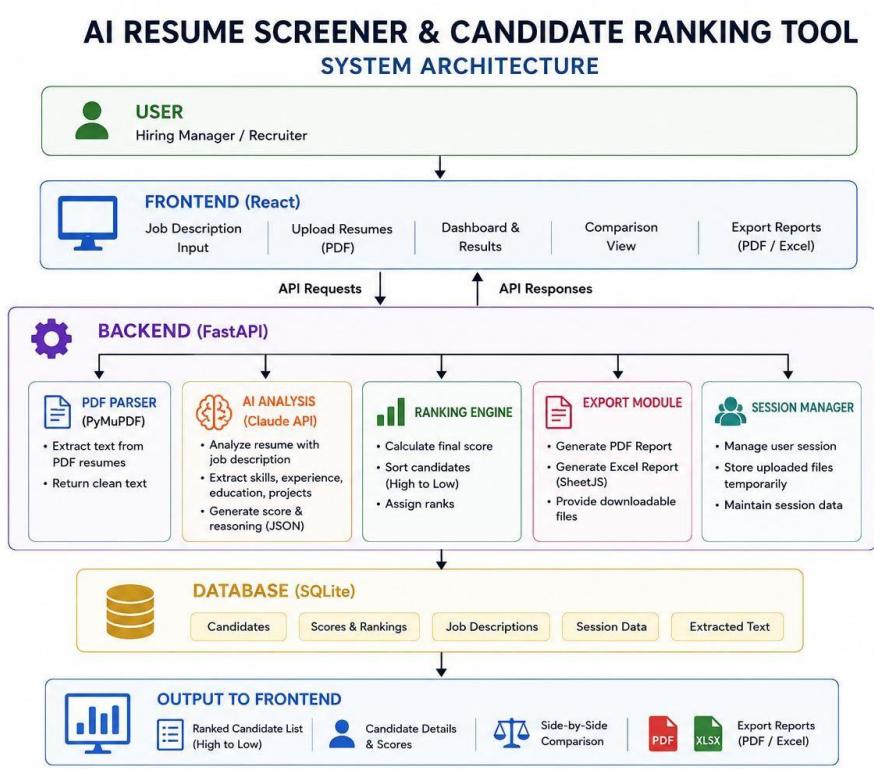
System Workflow

AI RESUME SCREENER & CANDIDATE RANKING TOOL

WORKFLOW DIAGRAM



System Architecture



Technology Stack

Frontend:	- React - Tailwind CSS
Backend:	FastAPI
PDF Parsing:	- PyMuPDF - pdfplumber
Artificial Intelligence:	Mistral API
Database:	SQLite
Export:	- Pandas Excel reports (.xlsx). - ReportLab PDF reports (.pdf).
Deployment:	- Vercel - Render

Team Roles --- Responsibilities

Harshini S - Frontend

- UI Design
- React Components
- User Experience
- Result Display

Konda Yashas Sree - Backend

- FastAPI Setup
- API Development
- Frontend-Backend Integration

Natarajan S - AI s Resume Processing

- API Workflow
- Resume Analysis Logic
- Resume Scoring Logic
- JSON Extraction Flow

Madduri Sai Mythili - PDF Parsing s Export

- PDF Text Extraction
- Export Module
- Integration Testing
- Documentation

AI Scoring System:

We have designed a weighted scoring system.



Candidates

HR Manager
Administrator

HR

Search by name or skill...

AllStrongly RecommendRecommendConsiderNot Recommended

CMP	RANK	CANDIDATE	SCORE	EXPERIENCE	SKILLS	STATUS
☐	1	Karthik Raj Resume_3.pdf	100	7 Years	JavaPython+28	Strongly Recommend
☐	2	Aishwarya Menon Email1.pdf	93	3 Years	PythonJavaScript+10	Strongly Recommend
☐	3	Rohit Sharma Email2.pdf	93	5 Years	JavaPython+18	Strongly Recommend
☐	4	Sneha Raj Resume_4.pdf	93	4 Years	JavaPython+23	Strongly Recommend
☐	5	Rahul Verma Resume_3.pdf	66	0.5 Years	JavaPython+20	Consider
☐	6	Priya Sharma Resume_2.pdf	52	—	JavaPython+9	Consider
☐	7	Arun Kumar Resume_1.pdf	28	—	CC+++3	Not Recommended

Rating System:

- 80-100 → Strong Match → Strongly Recommend
- 50-79 → Partial Match → Consider
- 0-49 → Weak Match → Not Recommended

Hidden Skill Inference

Aishwarya Menon

Strongly Recommend

✕

Email1.pdf

93

Rank #2
Rating: Strong Match
Experience: 3 Years

CONTACT

☐

aishwarya.menon@gmail.com

9876543210

AI SUMMARY

The candidate is an exceptional match for the core requirements of the role, particularly in FastAPI, React, and cloud deployment. Their 3 years of relevant experience and project work demonstrate strong technical and problem-solving skills. However, the resume lacks explicit mention of Java/C++ (though Python/JavaScript are covered), NoSQL databases, and preferred frameworks like Spring Boot or Django. The education is solid but not from a globally top-tier institution. The certifications are basic but sufficient. Overall, this is a high-confidence Elite Match with minor gaps that are unlikely to impact performance significantly.

MATCHED SKILLS

PythonJavaScriptReact.jsFastAPIREST APIs

SQL (MySQLPostgreSQL)GitDockerAWS (EC2S3)

Agile methodologies

MISSING SKILLS

JavaC++NoSQL databasesSpring BootDjango

AngularCloud platforms beyond AWS (e.g. GCPAzure)

AI/MLCybersecurity

Smarter Than Keyword Matching

Our AI doesn't just match keywords – it infers skills intelligently:

- 500+ LeetCode problems → **DSA** inferred
- Spring Boot experience → **Java Backend** inferred
- MERN Stack → **REST APIs** inferred
- AWS Certification → **Cloud Deployment** inferred

Deployment



Deployment

Live & Running

Component	Platform	Status
 Frontend	 Vercel	 Live
 Backend	 Render	 Live
 Database	 SQLite	 Active
 AI Engine	 Mistral API	 Connected

Output Achieved:

Harshini S- Frontend & UI/UX Lead

ResumeAI

Smart Hiring. Smarter Future.

Dashboard

Dashboard

Upload Resumes

Candidates

History

Reports

Analytics

Settings

About Us

Next-Gen AI Recruitment Intelligence

Smarter screening. Faster hiring. Deep analysis powered by Mistral Large 2411.

Start New Analysis

TOTAL CANDIDATES

7

AVG MATCH SCORE

75%

HIGHEST SCORE

100%

STRONG FITS

4

Recommended Candidates

1

Karthik Raj

7 Years

100%

Java

Python

Go

EXTREMELY RECOMMENDED

2

Aishwarya Menon

3 Years

95%

Python

JavaScript

React.js

EXTREMELY RECOMMENDED

3

Rohit Sharma

5 Years

93%

Java

Python

JavaScript

EXTREMELY RECOMMENDED

4

Sneha Raj

2 Years

90%

Python

JavaScript

React.js

EXTREMELY RECOMMENDED

FULL LEADERBOARD

...

AI Powered Recruitment
Powered by Mistral Large 2411

Harshini S- Frontend & UI/UX Lead

- ❖ **Core UI/UX Development:** Architected the entire application interface using React.js and Tailwind CSS, ensuring a responsive, component-based design for a seamless user experience.
- ❖ **Intelligent Input Modules:** Developed the Job Description (JD) Input and Multi-Resume Upload systems, featuring real-time validation, state management, and drag-and-drop PDF handling.
- ❖ **System Workflow & Navigation:** Engineered a multi-stage workflow visualization (Input → Analysis → Results) to guide recruiters through the screening process intuitively.
- ❖ **Integration Architecture:** Designed the frontend-to-backend service layer to communicate with the FastAPI/Mistral AI engine, managing complex request-response cycles and data flow.
- ❖ **Advanced Results Dashboard:** Conceptualized and built the results interface, featuring:
 - ❖ **Candidate Ranking Cards:** Visualizing scores, skills, and AI recommendations.
 - ❖ **Comparison Engine:** Side-by-side layouts for top-tier candidate evaluation.
 - ❖ **Data Export Suite:** Reporting structures for PDF and Excel data extraction.
- ❖ **Testing & Optimization:** Performed rigorous cross-device testing using Vite, ensuring 100% responsiveness and smooth interaction logic before final deployment.
- ❖ **Cross-Functional Collaboration:** Defined the API data structures and coordinated with the AI/Backend teams to bridge the gap between raw screening data and a polished user dashboard.

Konda Yashas Sree - Backend & Database Architect

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/candidates' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/candidates

Server response

Code	Details
200	Response body <pre>[{ "id": 1, "name": "Arun Kumar", "score": 87 }, { "id": 2, "name": "Rahul", "score": 76 }]</pre> Download

Response headers

```
content-length: 76
content-type: application/json
date: Thu, 11 Jun 2026 07:51:03 GMT
server: uvicorn
```

Responses

Code	Description	Links
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AI Resume Screener Backend 0.1.0 OAS 3.1

/openapi.json

default

POST

/upload

Upload Resume

▼

GET

/candidates

Get Candidates

▼

GET

/rankings

Get Rankings

▼

GET

/job-description

Get Job Description

▼

POST

/job-description

Save Job Description

▼

GET

/

Root

▼

GET

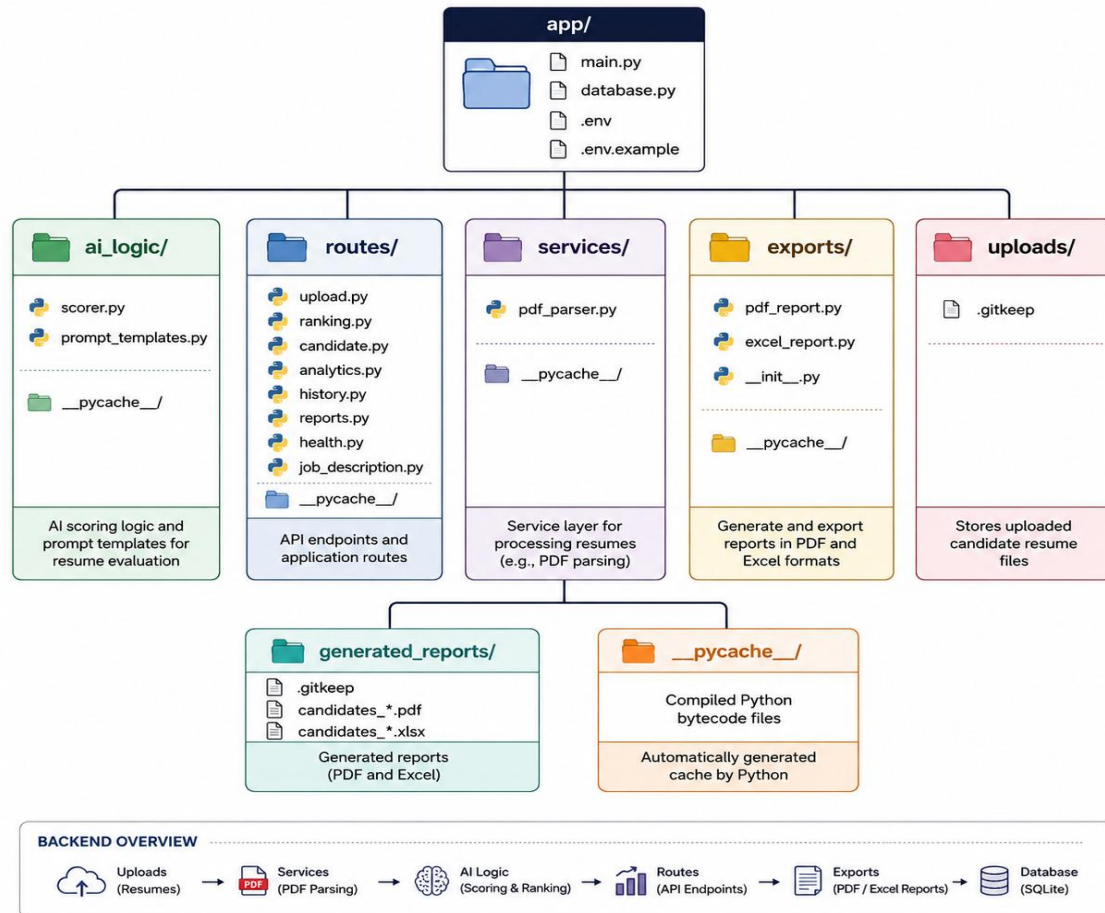
/health

Health Check

▼

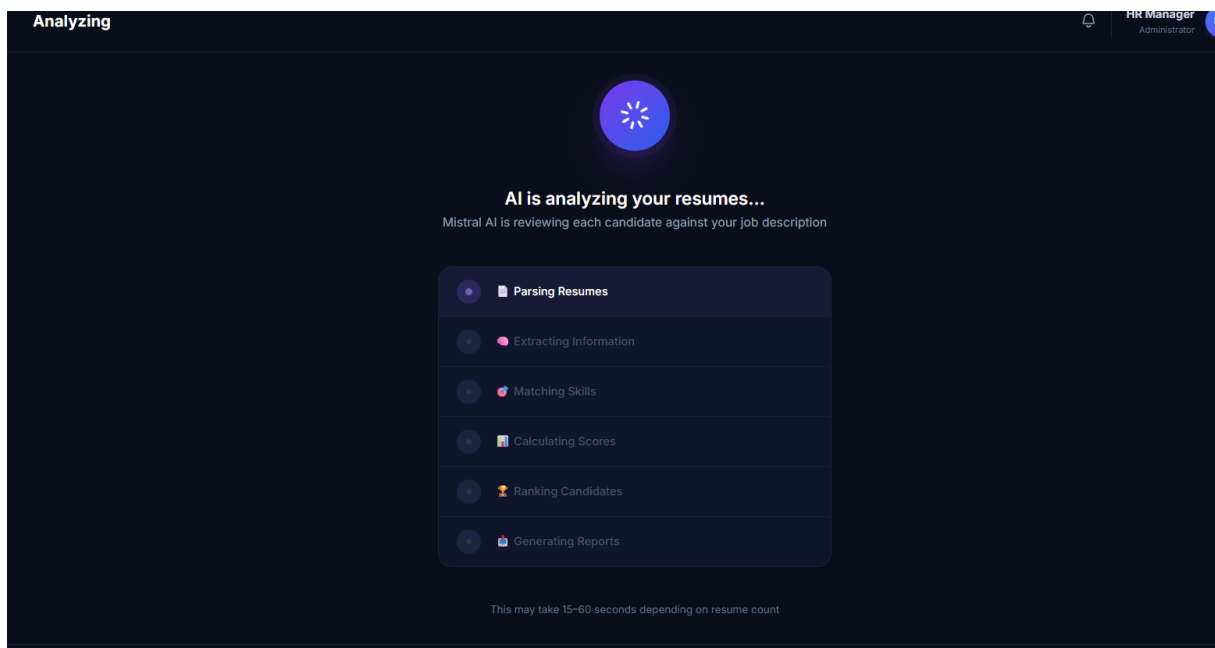
Schemas

AI-RESUME-SCREENER BACKEND STRUCTURE



- ❖ **FastAPI Backend Architecture:** Engineered the project's core server logic using FastAPI, developing high-performance RESTful API endpoints for candidate management, ranking results, and job descriptions.
- ❖ **Database Design & ORM:** Architected a robust relational data model using SQLite and SQLAlchemy, implementing secure storage and session management for candidate profiles and __pycache__ contents.
- ❖ **Data Pipeline Integration:** Seamlessly bridged the gap between raw file uploads and database persistence, ensuring the AI scoring engine could dynamically retrieve and process candidate information.
- ❖ **API Orchestration & Validation:** Configured Swagger UI for comprehensive API documentation and performed rigorous backend testing to ensure 100% reliability in data transmission between modules.
- ❖ **System Collaboration:** Played a key role in the full-stack integration, synchronizing database models with the frontend requirements and AI processing logic.

Natarajan - AI & Systems Integration Lead



```
C:\Users\natar\AI-Resume-Screener\backend>python -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\natar\\AI-Resume-Screener\\backend']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [23516] using WatchFiles

MISTRAL BATCH-PROCESSOR: ONLINE
✓ Model: mistral-large-2411 (High Speed)
✓ Status: API Key Detected

SKIP SEEDING: No User model found in database.py or models.py
INFO: Started server process [17692]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

- ❖ **Mistral AI Core Implementation:** Architected the intelligent screening engine using Mistral-small-latest, optimizing for high-speed candidate analysis and precise reasoning.
- ❖ **Intelligent Scoring Algorithm:** Developed a weighted scoring logic that evaluates candidates across multiple dimensions, including technical skill density, experience relevance, and role fit.
- ❖ **System Robustness & Scale:** Built a specialized Safe-JSON Parser and Rate-Limit Handler (429 detection) to ensure the system remains stable during high-volume batch resume processing.
- ❖ **API Orchestration:** Engineered the seamless integration between the Mistral LLM and the FastAPI backend, ensuring structured, production-ready data flow.
- ❖ **Lead System Integrator** Spearheaded the final project clubbing phase—merging the individual AI, Backend, and Frontend modules into a unified, deployed repository.



Name: Karthik Raj

Expe

Rank: 2 | Score: 91/100 | Recommendation: Strongly Recommend

Name: Sneha Raj

Experience: 4 Years -----

Rank: 3 | Score: 70/100 | Recommendation: Consider

Name: Rahul Verma

Experience: 0.5 Years -----

Rank: 4 | Score: 50/100 | Recommendation: Consider

Name: Priya Sharma

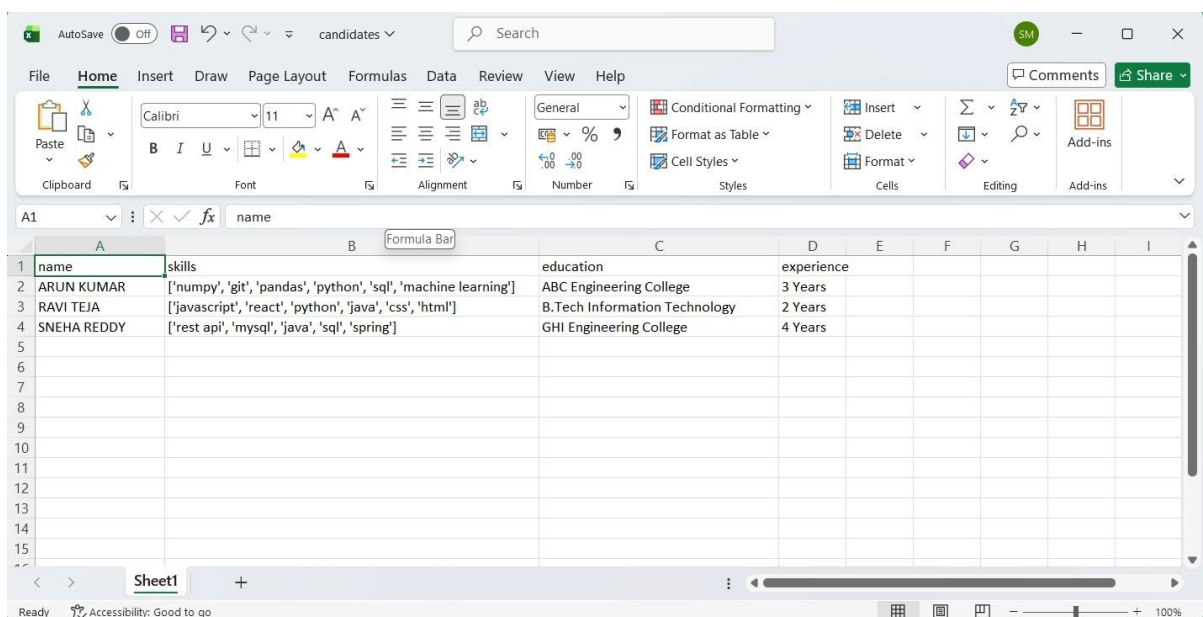
Experience: 0.25 Years -----

Rank: 5 | Score: 35/100 | Recommendation: Not Recommended

Name: Arun Kumar

Skills: C++, Basic Java, HTML, CSS, Basic SQL

Experience: —



```
PS D:\AIRESUMESCREENER> python main_struct.py
```

```
=====
```

```
FILE: Resume1.pdf
```

```
=====
```

```
{'name': 'ARUN KUMAR', 'skills': ['machine learning', 'pandas', 'git', 'numpy', 'sql', 'python'], 'education': 'ABC Engineering College', 'experience': '3 Years'}
```

```
=====
```

```
FILE: Resume2.pdf
```

```
=====
```

```
{'name': 'RAVI TEJA', 'skills': ['java', 'css', 'html', 'react', 'python', 'javascript'], 'education': 'B.Tech Information Technology', 'experience': '2 Years'}
```

```
=====
```

```
FILE: Resume3.pdf
```

```
=====
```

```
{'name': 'SNEHA REDDY', 'skills': ['java', 'mysql', 'spring', 'rest api', 'sql'], 'education': 'GHI Engineering College', 'experience': '4 Years'}
```

```
PS D:\AIRESUMESCREENER>
```

- ❖ **PDF Extraction Engine:** Engineered the project's core data ingestion layer using PyMuPDF, enabling high-fidelity text extraction from complex and diverse resume layouts.
 - ❖ **Data Pipeline Optimization:** Designed a specialized text cleaning pipeline to preserve critical sections like Skills, Experience, and Education, ensuring the AI received high-quality, structured input.
 - ❖ **Validation & Testing Suite:** Spearheaded the end-to-end testing phase by developing and executing validation scripts (main_ai_test.py, run_demo.py) to verify scoring accuracy and ranking reliability.
 - ❖ **AI Integration Support:** Collaborated closely with the AI module team to synchronize parser outputs with Mistral LLM requirements, guaranteeing seamless compatibility between modules.
 - ❖ **Technical Documentation:** Authored the comprehensive technical documentation for the extraction workflow, module usage, and integration procedures to ensure project scalability.
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Challenges Solutions

CHALLENGES & SOLUTIONS

 CHALLENGE	 SOLUTION
 API rate limits causing slow processing	 Switched to mistral-small-latest, added retry logic
 PDF text extraction inconsistency	 Used PyMuPDF with fallback to pdfplumber
 Frontend-backend data mismatch	 Standardized JSON schema across all modules
 Gemini quota blocked in India	 Migrated to Mistral API — no card required
 Free tier speed limitations	 Reduced model size + optimized delay timings

Conclusion

What We Achieved:

- Built a fully functional AI-powered resume screening tool
- Deployed live on Vercel and Render
- Real Mistral AI integration with intelligent scoring
- Complete team collaboration using GitHub
- End-to-end pipeline from PDF upload to ranked results

Impact:

- Reduces resume screening time from hours to seconds
- Removes human bias from initial shortlisting
- Makes professional hiring tools accessible to startups

"From 100 resumes to top 3 candidates — in under 60 seconds."

GitHub Repository

- Repository Name: AI-Resume-Screener
- Repository Link: [natarajannetworks/AI-Resume-Screener: AI Resume Screener and Candidate Ranking Tool using React, FastAPI and Claude API](https://github.com/natarajannetworks/AI-Resume-Screener)

Live Project Demonstration

Click the link above to view the live application.

Web URL: ai-resume-screener-fawn-two.vercel.app

Platform: Hosted on Vercel[1][2]

QR Code



“ Thank You!! ”

"Transforming recruitment through Artificial Intelligence."